

lab07

Lucas Rocha

```
library(DBI)
library(RSQLite)
conn <- dbConnect(RSQLite::SQLite(), "database.sqlite3")
```

1. Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística

```
SELECT DISTINCT i.id, i.name
FROM instructors i
JOIN teachings t ON i.id = t.instructor_id
JOIN sections sec ON t.section_uuid = sec.uuid
JOIN course_offerings co ON sec.course_offering_uuid = co.uuid
JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
JOIN subjects s ON sm.subject_code = s.code
WHERE s.abbreviation = 'STAT'
ORDER BY i.name;
```

id	name
4787911	ABIGAIL BENZINE
3111842	ADIN-CRISTIAN ANDREI
3775809	AKICHIKA OZEKI
4958796	ALAN HUANG
5451706	ALBERTO DEL PIA
3661782	ALBRECHT KLEMM
5656034	ALEXANDER COVINGTON
4680082	ALEXANDER FISH
3573775	ALEXANDER KISELEV
5162339	ALYSSA DIGILIO

R:1 Total sao 379 professores

2.O GPA americano é definido numa escala de 0 a 4, em que A = 4, AB = 3.5, B = 3, BC = 2.5, C = 2, D = 1 e F = 0. Determinando a nota média de cada oferecimento pela ponderação da quantidade de

alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística): Quem é o professor mais difícil? Quem é o professor mais fácil? Qual é a disciplina mais difícil? Qual é a disciplina mais fácil?

```
WITH gpa_professores AS (
    SELECT
        i.name AS nome,
        SUM(g.a_count * 4.0 + g.ab_count * 3.5 + g.b_count * 3.0 + g.bc_count *
2.5 + g.c_count * 2.0 + g.d_count * 1.0) /
        CAST(SUM(g.a_count + g.ab_count + g.b_count + g.bc_count + g.c_count +
g.d_count + g.f_count) AS FLOAT) AS gpa_medio
        FROM grade_distributions g
        JOIN sections sec ON g.course_offering_uuid = sec.course_offering_uuid AND
g.section_number = sec.number
        JOIN teachings t ON sec.uuid = t.section_uuid
        JOIN instructors i ON t.instructor_id = i.id
            JOIN subject_memberships sm ON g.course_offering_uuid =
sm.course_offering_uuid
        JOIN subjects s ON sm.subject_code = s.code
        WHERE s.abbreviation = 'STAT'
        GROUP BY i.id, i.name
        HAVING SUM(g.a_count + g.ab_count + g.b_count + g.bc_count + g.c_count +
g.d_count + g.f_count) > 0
    ),
gpa_disciplinas AS (
    SELECT
        c.name AS nome,
        SUM(g.a_count * 4.0 + g.ab_count * 3.5 + g.b_count * 3.0 + g.bc_count *
2.5 + g.c_count * 2.0 + g.d_count * 1.0) /
        CAST(SUM(g.a_count + g.ab_count + g.b_count + g.bc_count + g.c_count +
g.d_count + g.f_count) AS FLOAT) AS gpa_medio
        FROM grade_distributions g
        JOIN course_offerings co ON g.course_offering_uuid = co.uuid
        JOIN courses c ON co.course_uuid = c.uuid
        JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
        JOIN subjects s ON sm.subject_code = s.code
        WHERE s.abbreviation = 'STAT'
        GROUP BY c.uuid, c.name
        HAVING SUM(g.a_count + g.ab_count + g.b_count + g.bc_count + g.c_count +
g.d_count + g.f_count) > 0
    )

SELECT * FROM (SELECT 'Prof. Mais Difícil' AS tipo, nome, gpa_medio FROM
gpa_professores ORDER BY gpa_medio ASC LIMIT 1)
```

```

UNION ALL
SELECT * FROM (SELECT 'Prof. Mais Fácil' AS tipo, nome, gpa_medio FROM
gpa_professores ORDER BY gpa_medio DESC LIMIT 1)
UNION ALL
SELECT * FROM (SELECT 'Disc. Mais Difícil' AS tipo, nome, gpa_medio FROM
gpa_disciplinas ORDER BY gpa_medio ASC LIMIT 1)
UNION ALL
SELECT * FROM (SELECT 'Disc. Mais Fácil' AS tipo, nome, gpa_medio FROM
gpa_disciplinas ORDER BY gpa_medio DESC LIMIT 1);

```

tipo	nome	gpa_medio
Prof. Mais Difícil	JAMES D KUELBS	2.637500
Prof. Mais Fácil	SRIKANTHMADHAVAN ARAVAMUTHAN	4.000000
Disc. Mais Difícil	Introduction to the Theory of Probability	2.886374
Disc. Mais Fácil	Nonparametric Statistics and Machine Learning Methods	4.000000

```

# Desconecta do banco de dados SQLite
dbDisconnect(conn)

```

```

Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()

```

```

[1] "2026-09-23 21:05:48 -03"

```